Discover Applied Sciences

Research

Side-channel attacks on convolutional neural networks based on the hybrid attention mechanism

Tao Feng[1] · **Huan Gao**[1] · **Xiaomin Li**[2] · **Chunyan Liu**[3]

Received: 21 December 2024 / Accepted: 3 April 2025
Published online: 24 April 2025
© The Author(s) 2025 OPEN

Abstract
In the field of security assessment of password chips, side-channel attacks are an important and effective means of extracting sensitive information by analysing the physical characteristics of the chip during operation, providing an important basis for security assessment. In recent years, deep learning technology has been widely used in the field of side-channel attacks, which can automatically learn and identify the physical leakage characteristics of the chip and improve the efficiency and accuracy of the attack. However, deep learning-based side-channel attacks may be disturbed by environmental noise during the training process, and there are also problems of model overfitting and slow convergence. In order to more effectively extract feature information in the trajectory to implement a side-channel attack, this paper proposes a new attention mechanism convolutional neural network model architecture. The model combines a convolutional neural network with an attention mechanism. It optimises the traditional CNN model by improving the convolutional layer and introducing a fused hybrid attention mechanism, enhancing the model's ability to capture global information to effectively extract relevant leaked information. Experimental results show that the model has good attack results on the ASCAD public dataset. Compared with other models, it requires 74.87% less power consumption for side-channel analysis, and the model accuracy is significantly improved. It solves the problems of overfitting and slow convergence speed, and can meet the requirements of side-channel modeling and analysis.

Article Highlights

- Design an efficient convolutional neural network architecture model with an integrated attention mechanism.
- Improve the CBAM module and optimize the network structure for side-channel attacks.
- Significantly improve the convergence speed and attack efficiency of the neural network side channel model.

Keywords Side-channel attack · Deep learning · Convolutional neural network · Attention mechanism · Profiling attack

✉ Huan Gao, gh13572492218@163.com | [1]School of Computer and Communication, Lanzhou University of Technology, Lanzhou 730050, China. [2]Zhejiang Scientific Research Institute of Transport, Zhejiang 310006, China. [3]Lanzhou University of Technology, Lanzhou 730050, China.

1 Introduction

With the rapid development of computer technology such as the Internet, cloud computing, and big data, the demand for the transmission, storage, and processing of digital data is constantly increasing, and the security of embedded devices is particularly important [1–3]. As an important part of embedded devices, security chips are the main carrier of information security, and the security of the cryptographic algorithms running on them has been rigorously verified mathematically [4–8]. However, during the actual operation of cryptographic devices, it is inevitable that physical information related to sensitive intermediate values will be leaked, such as operating time, energy consumption, electromagnetic radiation, etc., which are often referred to as side-channel information [9]. Kocher et al. [10] first proposed the Side Channel Attack (SCA), which uses the correlation between side-channel information and the internal operating data of the cryptographic algorithm, combined with statistical tools to crack the encryption algorithm key with a small computational cost. Side channel attacks are divided into Profiling Attack (PA) and Non-profiling attack (NPA) [11]. NPA use statistical analysis tools to calculate the correlation between the leak point and sensitive data, which mainly include simple power analysis, correlation power analysis and differential power analysis. PA mainly include template attacks and machine learning attacks, which mainly depict the key characteristics, and the template is used to attack and recover the key of the target device [12].

In recent years, deep learning technology has opened up new technical paths for side-channel attacks due to its powerful feature extraction capabilities [13–15]. Compared with the traditional template attack, which requires manual design of a leakage model, the deep learning-based side-channel attack can automatically mine the nonlinear features in the power consumption energy trace [16]. By fusing the neural network with the side-channel attack technique, not only can the security of the encryption algorithm be bypassed at the mathematical theory level, but the attack capability on the encryption algorithm can also be improved. Even in encryption algorithms with defensive masks, the deep learning-based side-channel attack technique can still successfully attack with a small amount of energy trace curves, significantly improving the success rate of side-channel attacks and overcoming protection techniques such as masking, random delays and clock jitter [17]. Maghrebi et al. [18] were the first to apply deep learning to side-channel attacks, demonstrating the advantages of deep learning in both protected and unprotected encryption implementations, and opening up the research direction of using deep learning to attack.

At present, the side-channel attack technology based on deep learning models has formed a multi-dimensional exploration system, and has achieved diversified research results. Hospodar et al. [19] first used the least squares support vector machine (SVM) model for side-channel attacks on AES encryption, and achieved the combination of machine learning and side-channel analysis through the adjustment of hyperparameters. Timon et al. [20] applied deep learning technology to non-modeling side-channel attack technology and proposing a differential deep learning analysis method. The attack is targeted at side-channel sensitive data, and the translation invariance of the convolutional neural network is used to counter the asynchronous energy traces. Experiments have proved that this method is superior to traditional non-template attack methods. Benadjila et al. [21] conducted an in-depth exploration of this field, introducing the ASCAD database to provide a rich resource for training and testing deep learning models. This not only demonstrates the effectiveness of deep learning side-channel attack techniques, but also lays a solid foundation for subsequent research. Alam et al. [22] proposed a machine learning-based multi-layer detection technique for detecting micro-architectural side-channel attacks. Hasan et al. [23] proposed a novel deep learning-based side-channel attack framework designed to address the challenge of hyperparameter tuning in complex neural network architectures. The proposed model, implemented using multi-layer perceptron and convolutional neural networks, demonstrates superior accuracy and computational efficiency compared to traditional side-channel attack models. Ito et al. [24] solved the dataset imbalance problem in deep learning side-channel attacks. Kim et al. [25] studied the effect of noise regularization on the training of convolutional neural networks and found that adding artificial noise to the input trace can improve the performance of the neural network. These studies have demonstrated the great potential of deep learning models in side-channel attacks.

Picek et al. [26] systematically analyzed deep learning-based side-channel attacks, exploring related threat models, different stages of attacks, and future research directions and challenges. Benadjila et al. [27] used the VGG-16 neural network model as a starting point to study side-channel attacks, proposed a model for the convolutional neural network architecture used in side-channel attacks, and gave specific hyperparameter settings. However, due to the need for multiple layers of fixed convolutional kernels to capture the local temporal features of the energy trace, the attack efficiency is poor in specific cases. Zaid et al. [28] proposed a side-channel attack method for constructing efficient CNN architectures in terms of both attack efficiency and network complexity, and used some visualization

techniques to explain the role of hyperparameters in the feature selection stage, verifying the high efficiency of the side-channel attack. However, when faced with dynamic desynchronization, the static convolutional architecture cannot adaptively select features, resulting in a decrease in attack efficiency.

Hu et al. [29] proposed that an improved denoising autoencoder filtering method can be introduced to improve the energy trace signal-to-noise ratio. Ahmed et al. [30] applied convolutional neural networks (CNNs) to side-channel analysis and conducted experiments on systems using 8-bit memory AVR microcontrollers and AES-128 encryption algorithms. showing that the side-channel attack of the convolutional neural network on the power consumption trace with Gaussian noise has better attack efficiency and success rate. The CNN model achieved good results in the side-channel attack of the AES encryption algorithm with a mask, and by automatically extracting features, it reduced the dependence on manual feature engineering, thereby improving the efficiency and success rate of the attack. Do et al. [31] fine-tuned the proposed network by adjusting hyperparameters such as the number of hidden layers and activation functions, solved the hyperparameter selection problem of multilayer perceptron and convolutional neural networks, and achieved high-performance detection of SCA. Lu et al. [32] proposed an end-to-end analysis of a neural network architecture model consisting of an encoder, an attention mechanism, and a classifier, which can abandon the manual feature extraction step and use fine-grained feature extraction and a lightweight network structure to achieve side-channel attacks on algorithms with mask protection.

Existing work has demonstrated that neural network models can improve the efficiency of side-channel attacks [33]. Neural network performance can be improved through different deep learning techniques, but current neural network models still have problems such as overfitting, poor noise resistance, gradient disappearance, and slow convergence. It is also costly to train a model with better classification results. Therefore, in order to solve these problems and explore a more efficient deep learning neural network model for side channel attacks, this paper proposes a side channel attack architecture model for convolutional neural networks based on an attention mechanism. The core purpose is to construct an efficient convolutional neural network model by introducing an attention mechanism in side channel attacks, in order to achieve efficient key recovery technology for mask-based protection encryption algorithms. The new neural network model studied in this paper aims to overcome the challenges of noise robustness and mask protection in existing deep learning side channel attack methods, simplify the complexity of the neural network model, and achieve breakthroughs in terms of architectural innovation, model generalization ability and computational efficiency.

This paper focuses on the neural network side channel attack of the AES encryption algorithm. The main contributions are as follows:

1. Network architecture: This paper proposes a more efficient convolutional neural network architecture for side channel attacks, uses the attention mechanism for side channel attacks, implements side channel attacks on convolutional neural networks, and improves the overfitting problem of the model.
2. Introduction and improvement of the CBAM attention mechanism: A modified CBAM attention module is introduced into the deep learning-based side channel attack. A hybrid attention mechanism module is designed by adding a nonlinear feature mixing mechanism, which gives greater weight to the feature points in the power consumption curve, so that the network can better learn the power consumption curve with strong operation dependence and enhanced data dependence.
3. Evaluation of the new architecture model: A deep learning side-channel attack model based on the hybrid attention mechanism was constructed, which solves problems such as slow convergence, over fitting, and gradient vanishing. It also demonstrates good attack results on public datasets, verifying the usability and efficiency of the model in side-channel attacks.

2 Background

2.1 Advanced encryption standard

Advanced encryption standard (AES) is a symmetric encryption algorithm published by the National Institute of Standards and Technology of the United States. It is designed to replace the old Data Encryption Standard (DES) as the new generation of encryption standards [34]. The AES algorithm uses a fixed-length key for encryption and decryption. The key length can be 128 bits, 192 bits, and 256 bits. It uses a block cipher to divide the plaintext into fixed-length blocks, each block being 128 bits long. Each block is encrypted separately and then combined to form the complete ciphertext. The energy consumption of the AES encryption algorithm during operation depends on the intermediate values

executed by the algorithm. When the AES encryption algorithm is executed, the input plaintext is first XORed with the key, and then the main round of encryption operations is performed. AES-128 performs 10 rounds, and AES-256 performs 14 rounds [35]. This includes the SubByte, ShiftRow, MixColumn, and AddRoundKey stages. In the first sub-operation of each round, a S-box is defined for AES. Z is defined as the output of the S-box, which is the intermediate value of the AES encryption algorithm. p indicates the plaintext, and k indicates the key bytes that perform an exclusive OR operation on the plaintext p. The output of the S-box is defined as follows:

$$Z = SBox(p \oplus k) \tag{1}$$

Since the plaintext is known, the output of the S-Box is related to the key. Therefore, the output of the S-box is sensitive data that can be used as an attack point.

2.2 Deep learning based side channel attacks

Deep learning side channel analysis (DLSCA) uses neural networks for modelling, which is divided into two phases: the modelling phase and the attack phase, corresponding to the training phase and the testing phase in the field of deep learning [36].

Training phase: In the training phase, the power consumption profile T_{train} running on the cipher chip is captured by an oscilloscope, and the leakage model Z related to the real key of the power consumption trajectory is computed based on the a priori information of the encryption algorithm. This information is used to train a deep learning model that predicts the data labels based on the power consumption trajectory T_{train}. If there is a mismatch, the model parameters are re-tuned to finally obtain a model M that maps the power consumption data into sensitive information Z.

Test phase: Prediction is performed on the test dataset using the trained deep learning model M to guess the key information. The prediction probability of each key candidate is calculated using the maximum likelihood score and the key with the maximum likelihood score is considered to be the correct key. A side channel attack is successful if and when the predicted key happens to be the true key [37].

2.3 Convolutional neural network

Convolutional neural network (CNN) is a specific type of neural network in the field of deep learning, which effectively extracts feature information from data information by mimicking the working mechanism of the human visual system [38]. CNN is composed of an input layer, a convolutional layer, an activation function, a pooling layer, and a fully-connected layer, which can be simplified into a feature extraction part and classification part.

The convolutional layer is the core component of the CNN [39], where the input samples are multiplied element-by-element with filters of different sizes and summed to generate a feature map [40]. Each filter in the convolutional layer generates a feature map, so the convolutional layer can extract several different features by using multiple filters. The convolutional layer includes the Stride parameter and Padding parameter. The Stride defines the length of the step that the filter moves over the input data, while the Padding is used to fill in extra values around the boundaries of the input data to avoid losing information and to ensure that the size of the output feature map is similar to the size of the input data. By stacking convolutional layers multiple times, deep convolutional neural networks can be constructed to extract more advanced and abstract feature representations. Convolutional layers are also used interchangeably with nonlinear activation functions and pooling layers to enhance the feature representation and reduce the size of the feature map.

The pooling layer, whose input is the output of the convolutional layer, compresses the input feature data with the purpose of extracting the main features and simplifying the network parameters, and is generally divided into average pooling and maximum pooling. Global Max Pooling (GMP) selects the largest data in the selected region as data output, while Global Average Pooling (GAP) performs average pooling for each channel of the feature map and outputs the average value of each channel. The fully connected layer is to split the output tensor of the pooling layer into some vectors again, multiply it with the weight matrix under the activation function and add the bias value, and use the gradient descent method to continuously optimize the loss function to achieve the final classification effect.

3 Convolutional neural network side channel attack architecture

Convolutional neural networks can automatically extract various features of the input trajectory through different filters in the convolution operation. However, irrelevant points in the input trajectory not only fail to provide useful information, but also introduce additional noise, which invalidates some of the extracted features and thus weakens the performance of the CNN. Existing side-channel attack models for neural networks also suffer from overfitting and slow convergence. Therefore, in order to improve the effectiveness of side channel attacks and optimize the ability of convolutional neural networks in feature extraction, this paper proposes a hybrid attention mechanism-based convolutional neural network side channel attack model HACNN-SCA. The model is based on a convolutional neural network CNN and introduces a hybrid attention mechanism module to optimize the performance of the side channel attack. The overall structure of the HACNN-SCA model is shown in Fig. 1. The input to the proposed hybrid attention mechanism convolutional neural network architecture comes from the raw power consumption curve data, and the output is the corresponding label category. The new architecture is roughly divided into three modules: the convolutional module, the hybrid attention mechanism module, and the classification module.

3.1 Convolution module

The basic principle of the convolutional layer is to first learn the coarse features of the input energy trace, and then refine it layer by layer. The HACNN-SCA system architecture consists of three convolutional modules, each of which contains convolutional layers, batch normalization layers, pooling layers, etc. A convolution operation is performed after scanning the input power consumption curve, and a batch normalization layer is introduced to process the values after convolution. The processed data is then activated using a ReLu activation function to introduce a nonlinear operation. These operations enable the entire model to learn more features of the energy trace.

3.2 Hybrid attention

The Convolutional Block Attention Module (CBAM) is one of the classic studies proposed by Woo et al. [41] in 2018. CBAM can enhance the representation ability of features in the attention network, paying attention to the leakage information of the power consumption curve. And it has been proven that using both channel attention and spatial attention in side-channel attacks on neural network models can effectively improve the prediction accuracy and computational efficiency of the side-channel attack model. However, this attention mechanism still has certain limitations. When using average pooling, the feature map becomes blurry, which affects the selection of the leakage point of the power consumption curve of the encryption algorithm. Drawing on this model, this paper designs a hybrid attention mechanism model for

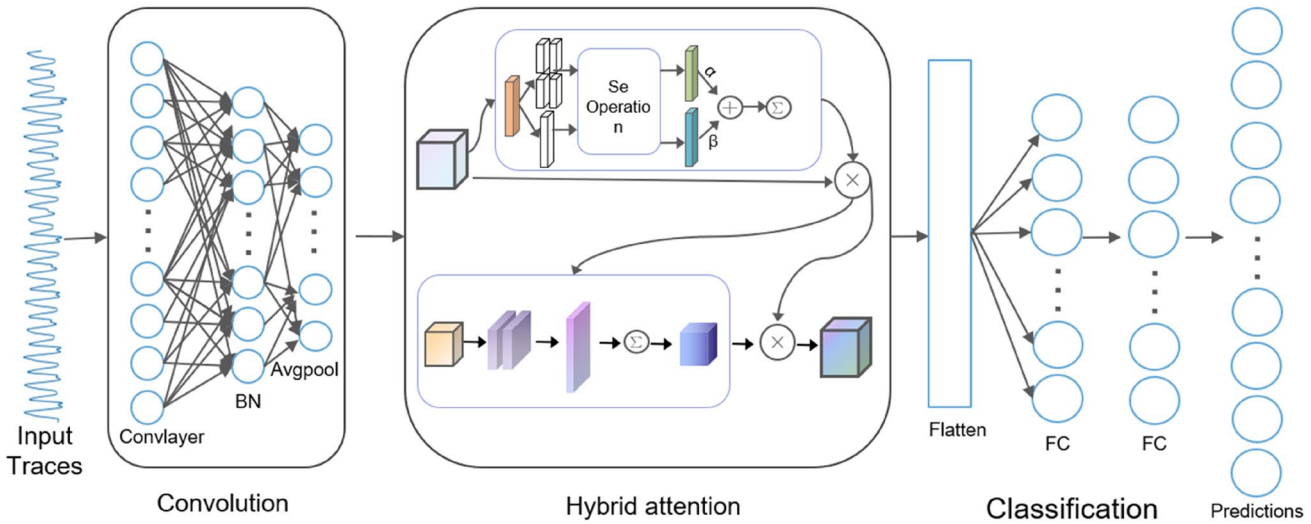


Fig. 1 HACNN-SCA model architecture

application in side-channel analysis. This model incorporates the SENet attention mechanism into the channel attention mechanism to form a new feature enhancement strategy. The improved hybrid attention feature fusion method rescales features in both the channel and spatial dimensions to eliminate the aliasing effect caused by feature fusion and focus on the salient parts of the input features.

The output of the first convolution result is used as the input of the attention mechanism module, the input feature is $x \in R^{C\times H\times W}$, and the result after channel convolution is $F_c \in R^{C\times 1\times 1}$. The output result of multiplying the convolved result with the input feature is used as the input to the spatial attention, and the final output is obtained by multiplying it with the original image after performing the $F_s \in R^{1\times H\times W}$ operation.

$$x' = F_c(x) \otimes x \tag{2}$$

$$x'' = F_s(x') \otimes x' \tag{3}$$

In Eqs. (2) and (3), the following variables are defined: $\otimes$ means that the input feature map is multiplied with the attention element one by one, x' is the feature map that has passed through the channel attention, and x'' denotes the final output.

In the Channel Attention Mechanism, the feature information contained in the side channel profile is adaptively selected by assigning weights to different features through the Channel attention module, and Spatial attention is used to point out the specific location of the features. As shown in Fig. 2, in the channel attention mechanism, the network uses a 1D Global Average Pooling operation (GAP) and a 1D Global Maximum Pooling operation (GMP) to obtain the maximum and average features in the 1D features in the power consumption curve, respectively. Meanwhile, the extracted weights are restructured by drawing on the compression and excitation mechanisms of SENet, and the restructured weights are finally weighted by adaptive weighting to obtain the final weights. In order to accelerate the convergence speed, the ReLU function is added to the compression and activation operations to filter out the negative values in the weights, and finally the Sigmoid function is used to normalize the data computation to generate the results of the channel weights, which facilitates the computation of the error and inversion. The channel attention module is specified in Eqs. (4) and (5) as follows.

$$F_c(x) = \sigma[SE(GAP(x)) * \alpha + SE(GMP(x)) * \beta] \otimes x \tag{4}$$

$$\alpha + \beta = \frac{1}{2} \tag{5}$$

where $F_c(x)$ is the output of the module,σ is the Sigmoid function, and α,β are the set hyperparameters for their respective weights, all following the rules of the above equation.

In the Spatial Attention mechanism, there is a discrepancy in the level of importance due to different locations in the feature map. To deal with this inconsistency, a spatial attention module is applied after the channel attention

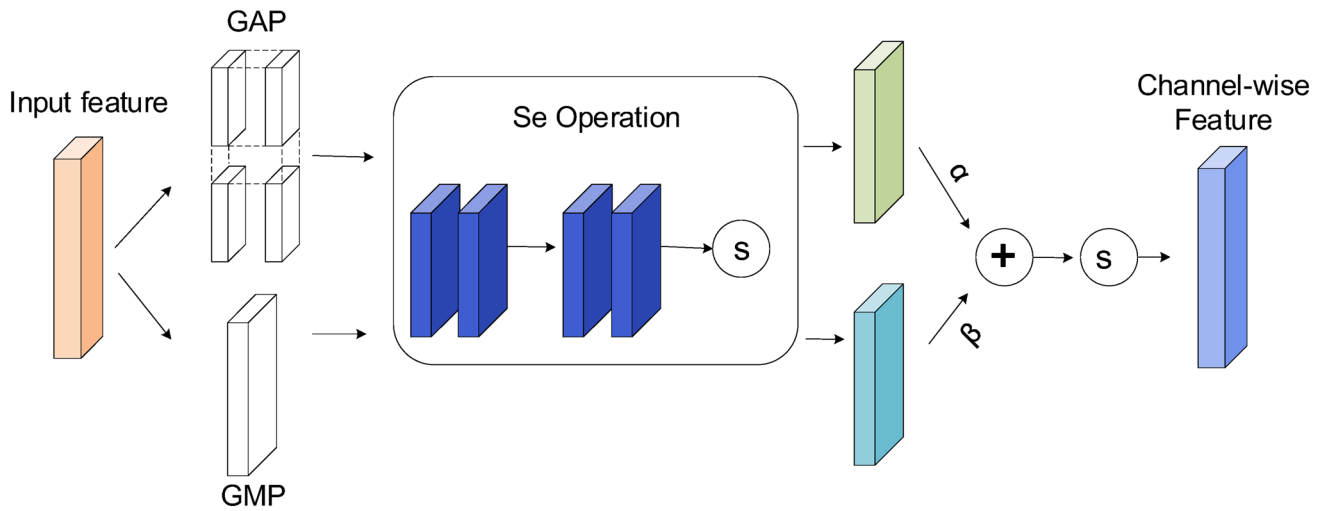


Fig. 2 Channel attention module

module to locate the important features. The Spatial Attention module is able to adaptively learn the weights of each spatial location based on the importance of the features to improve the model performance. The structure of the spatial attention module in CBAM is shown in Fig. 3. The module takes the feature map output from the channel attention module as the input feature map. Firstly, channel dimension compression is performed by performing global maximum pooling and global average pooling operations on the input feature map to obtain two feature matrices of size $W \times H \times 1$. Secondly, these two feature matrices are spliced in channel dimension and trained by convolutional neural network to obtain a training feature matrix. The spatial weight information is then generated through the Sigmoid activation function output to perform the normalization operation. The obtained spatial attention weights are multiplied element-by-element with the input feature matrix to weight the spatial locations of the intermediate features to obtain the final feature map. Finally, the attention module is inserted into the convolution block and the channel attention module and the spatial attention module are used sequentially. By introducing the improved CBAM module, the attention network is able to focus on the feature information points in the trajectory and reduce the noise introduced by non-essential points.

3.3 Classification module

The final setting of the HACNN-SCA model architecture is the classification module, which is used to classify the extracted feature information and predict the final result. The classification module is composed of a fully connected network and a SoftMax activation function. At the same time, in order to alleviate the overfitting problem that occurs during model training, a Dropout layer is introduced after the fully connected layer. This module consists of two fully connected networks. After passing through the Dropout layer, SoftMax generates the most likely key probability distribution, and the most likely key prediction result is selected based on the network output.

3.4 Implementation of a convolutional neural network side channel attack

In the side channel attack on convolutional neural networks based on the hybrid attention mechanism, deep learning and attention mechanisms are mainly used to improve the efficiency and accuracy of the attack. The attack process of this architecture model is as follows: after initializing the neural network parameters, the HACNN network is trained according to the label of each key guess. The attention module is used to extract the feature information of the power consumption data, and the standard deviation score of the attention module is recorded during each iteration, as well as the ranking of the current guess key. The final attack effect is calculated based on the data collected during the attack process, and the final average key ranking is returned. The specific process is shown in Algorithm 1, which summarizes the training process of the improved attention mechanism side channel attack model.

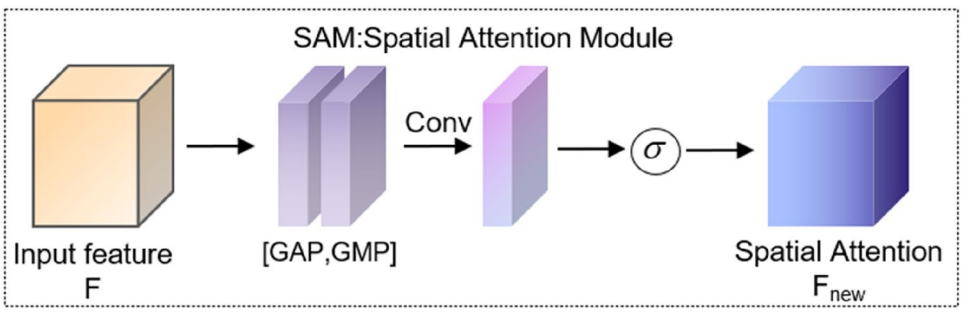


Fig. 3 Spatial attention module

Algorithm 1 Improved side channel attacker based on attention mechanism

Inputs: N power traces T_i,
corresponding plaintexts, substitution box SBox
Output: keyRank: Average key rank of one byte
1) function define the HACNN network Net and parameters
2) return HACNN Net
3) **for** $k \hat{I}$ (0,255) **do**
4) Compute the hypothetical intermediate values $(Z_{i,k})1 £ i £ N = SBox(P_i Å k)$
5) Set training labels as $Y = (Z_{i,k})_{(0£i£N)}$
6) Initialize the network parameters of HACNN
7) **for** $e < N_e$ **do**
8) Features=Extract Features (using CBAM)
9) Perform Deep Learning Training Process: $DL_k = HACNN\{T_i, Y\}$
10) Calculating and preserving the standard deviation of hybrid attention
11) Record the rank score
12) **end for**
13) Compute metric m_k according to (4) and (6)
14) **end for**
15) $M_k = \max(m_k)$
16) return rank

4 Experiment

4.1 Experimental configuration

The experimental implementation in this paper is based on the Keras deep learning open source library, and the model training runs on a CPU configured with an Intel(R) Xeon(R) CPU E5-2680 v4 and a GPU with RTX4060-8G. All experiments are implemented in the Python programming language, using the Tensorflow 2.10.0 framework and Cuda 11.2.2 for model training, and the ASCAD public dataset is used for the experiments.

ASCAD is an experimental platform introduced by Benadjila et al. [21]. The ASCAD dataset is collected based on the AES-128 encryption algorithm implemented on an ATMega8515 microcontroller, and generated by capturing the electromagnetic radiation signals generated when the smart card performs encryption operations. This dataset is designed to evaluate the effectiveness of side-channel attacks based on machine learning and deep learning, and is a widely used benchmark dataset. The ASCAD dataset contains 50,000 training energy traces and 10,000 test energy traces. The samples corresponding to the first round of the third S-box leakage operation during the AES-128 execution process are pre-selected, and each energy trace has 700 sample time points. The leakage model is shown in Eq. (6). The ASCAD dataset contains three h5 files: ASCAD.h5, ASCAD desync50.h5, and ASCAD desync100.h5. The ASCAD.h5 file contains the dataset synchronized without jitter. ASCAD desync50.h5 file contains the jitter of 50 sample windows. The electromagnetic energy trace in ASCAD desync100.h5 is based on ASCAD.h5 with a random offset of 100 time points. ASCAD dataset is available at https://github.com/ANSSI-FR/ASCAD.

$$Z(k^*) = Sbox(P_3 \oplus k^*) \tag{6}$$

Figure 4 shows an example of multiple curves from the ASCAD dataset, providing a visual representation of side-channel power consumption waveforms. The leakage characteristics of power traces with offsets of 0, 50, and 100 can be

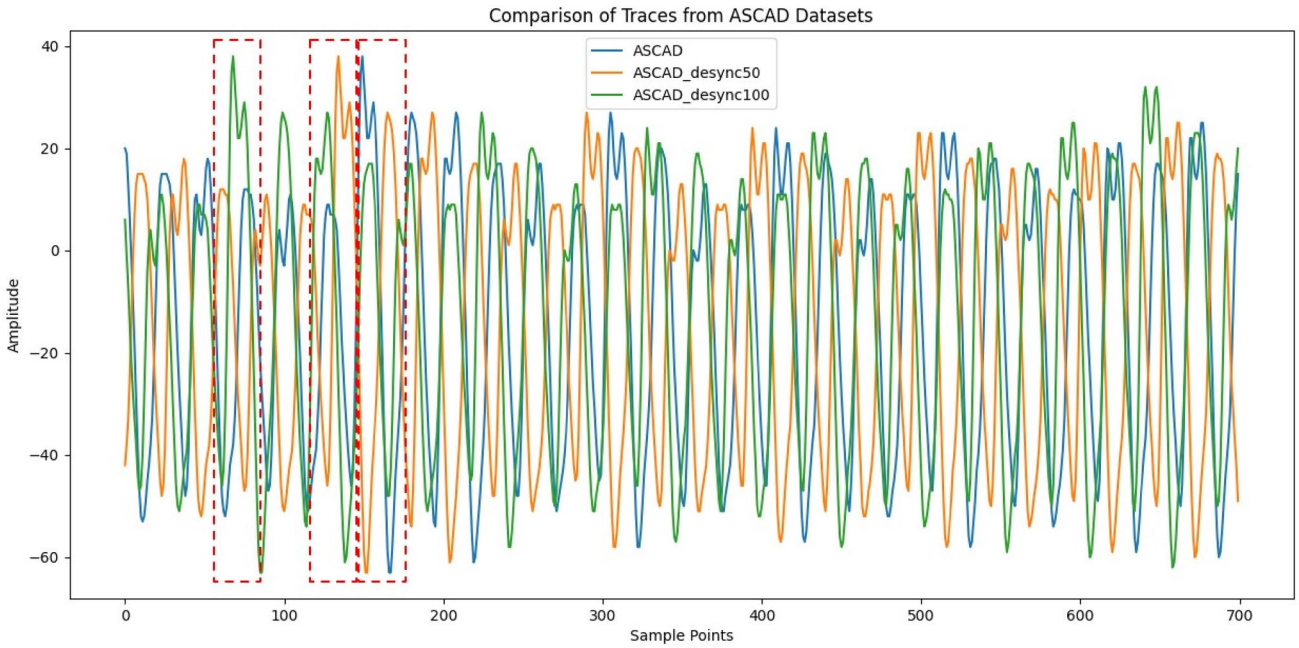


Fig. 4 Example of ASCAD power consumption curve

clearly observed. In the graph, the horizontal axis represents time points, while the vertical axis indicates the amplitude of electromagnetic energy.

4.2 Evaluation metrics

Rank and Guess Entropy (GE) are commonly used side-channel analysis evaluation metrics when assessing the effectiveness of side-channel attacks. In this paper, the rank function and accuracy rate are used to evaluate the performance of the model.

4.2.1 Rank

The key guessing vector is denoted as $g = \{g_1, g_2, \ldots, g_{|K|}\}$ and the number of attack measurements is denoted as N_a, where the keys are arranged in descending order of predicted probability, $|K|$ denotes the size of the key space, and k^* denotes the key for the encryption algorithm [42]. A different candidate key, all of which are distinguished using the maximum log-likelihood distinguisher, will result in a candidate key probability score g_i. g_i is specifically represented by Eq. (7):

$$g_i = \sum_{i=1}^{N_a} \log(\hat{p}_{ij}) \tag{7}$$

where $\hat{p}_{ij}$ denotes the predicted probability of the ith key candidate in the jth attack measurement. The rank denotes the average position of k^* in g. When the rank is equal to 0, this means that k^* is equivalent to g_1, indicating that the attack is successful. The goal of the final attack is to achieve a successful attack using the minimum number of attack traces.

The corresponding rank function for a model trained using dataset D_{train} and tested using dataset D_{test} is defined as follows:

$$rank(D_{train}, D_{test}, n) = \left|\left\{k \in K \middle| \vec{g}_n[k] > \vec{g}_n[k^*]\right|\right\} \tag{8}$$

In Eq. (8), $\vec{g}_n[k]$ is denoted as the probability vector of the candidate key and $\vec{g}_n[k^*]$ is the probability vector in the case of a known real key.

4.2.2 Accuracy

Accuracy is a generic metric for classification problems in deep learning and refers to the proportion of correctly predicted classification results to the total samples [43]. The probability vectors output from the model are processed using reverse solo thermal coding, i.e., the categories are represented by finding the subscript corresponding to the maximum value in the probability vector. The output probability vector of the classification model is compared with the validation set and the number of samples that are correctly classified is counted and divided by the total number of samples, i.e., the accuracy definition is obtained as shown in Eq. (9):

$$acc = \frac{\left|\left\{(\vec{I}_i, p_i, k^*) \in D_{test} \middle| k^* = \arg\max_{k \in K} \vec{y}_i[k]\right|\right\}}{|D_{test}|} \tag{9}$$

4.3 Experimental results and analysis

Although deep learning models usually exist as black boxes with relatively low interpretability, their performance can still be optimized by training to select the hyperparameters of the neural network model to achieve the best results. To evaluate the effectiveness of the model in this paper, experiments were conducted on the ASCAD dataset, which provides three different datasets with asynchronous values. The training energy trace was 50,000, the test energy trace was 10,000, the dimension of the input power consumption curve is (None, 700, 1) and the output is 256. The optimizer is Adam, the activation function used after the convolutional layer is ReLu, and batch normalization is used to prevent overfitting and accelerate the model. The batch size is set to 50, which is optimal, and the learning rate is set to 0.00001. In the experiments of this paper, multiple attack tests are con conducted against the proposed neural network model. Given the large impact of hyperparameters on the experimental performance, this paper provides an in-depth analysis of the impact of batch size on the correct key guessing results under the condition of varying number of trajectories used for attacks. Table 1 demonstrates the network traversal optimization search range for specific hyperparameters as well as the final hyperparameter selection results.

4.3.1 Ablation experiment

In order to construct a more effective analytical model and illustrate the advantages of the proposed hybrid attention mechanism over traditional neural network models, this paper integrates the proposed hybrid attention mechanism module and SeNet module into the standard CNN described above. In this section of the experiment, ablation experiments are used to systematically explore the impact of different schemes on the efficiency of the model.

For the ASCAD dataset synchronous trajectory, the accuracy of attack success varies with different processing methods. Therefore, we set up the following experiments for comparative analysis: (A) the benchmark model with only CNN, (B) SeNet, and (C) the hybrid attention mechanism module.

Table 2 shows the performance results of each model, which verifies that the introduction of a hybrid attention mechanism module into the convolutional neural network can effectively extract power consumption curve information. As can be seen from Table 2, when only the CNN baseline model (C) is used for side-channel attacks, the number of power consumption curves required for the rank to approach 0 after 100 model iterations is 286, and the model accuracy is 89.14%. However, when an excitation module is added to the CNN benchmark model, i.e. the A + B module, it is shown that 165 power consumption curves are required after about 80 model iterations. Compared with the side-channel attack using the CNN benchmark model, the number of curves required is reduced by 42.31%, the number of iterations

Table 1 Hyperparameter selection range and final choice

Hyperparameter	Select range	Select
Optimizer	Adam, RMSProb, SGD	Adam
Activation function	Tanh, Relu, SeLU	ReLu
Learning rate	0.00001–0.1	0.00001
Batch size	32, 50, 64, 128, 256, 512	50

Table 2 Performance comparison of side channel attack models

Method	Traces	Epoch	Average accuracy (%)
A	286	60	89.14
A + B	165	80	94.31
A + C	48	100	98.12

is shortened, the accuracy rate is improved to 94.31%, and the model convergence speed is accelerated. Secondly, when a hybrid attention mechanism, i.e. the A + C module, is added to the benchmark CNN model, it takes about 60 iterations and 48 power consumption curves to successfully attack, with a model accuracy of 98.12%. Compared with other combined modules, the convergence speed is faster. Compared with the benchmark CNN model, the number of power consumption curves is reduced by 83.22%. Compared with the addition of the SeNet module, the number of power consumption curves is reduced by 70.91%.

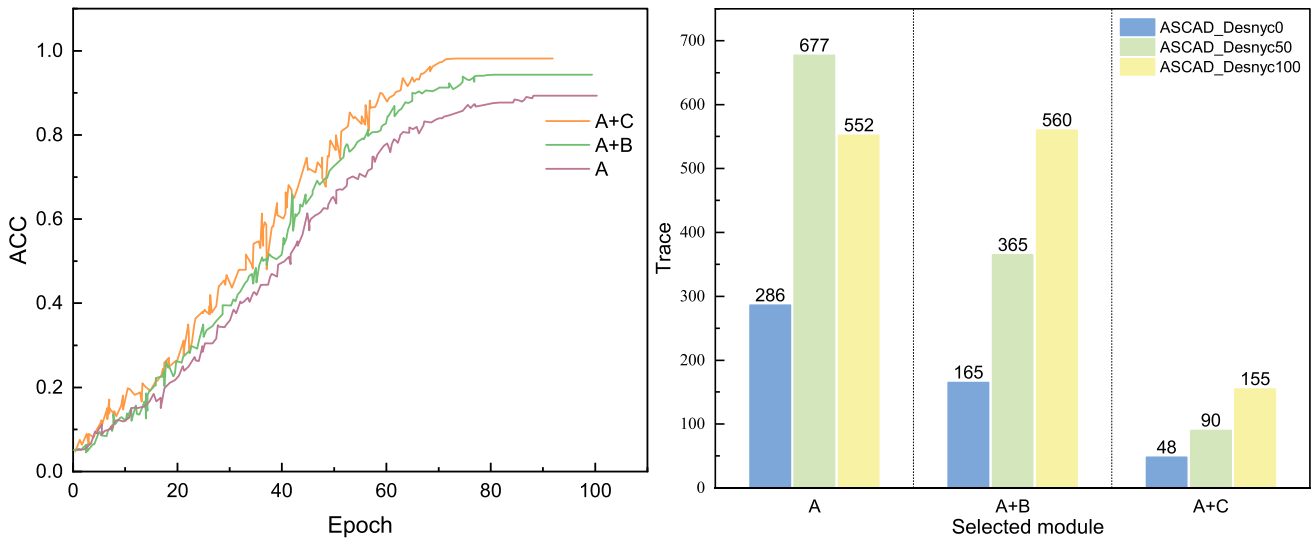
As shown in Fig. 5a, which details the iteration period and accuracy of each module, it can be seen that the hybrid attention mechanism is superior to other combined models. The hybrid attention mechanism proposed in this paper has higher accuracy and faster convergence, significantly improving the performance of the side-channel attack model. However, after the model converges faster, its curve has large fluctuations.

To further validate the effectiveness of the hybrid attention mechanism, the A + C module is integrated into the baseline convolutional neural network for side channel analysis and experiments are conducted on three datasets with different asynchronous values. The following conclusions are drawn after comparative analysis, and it can be seen from Fig. 5b that the hybrid attention mechanism in this paper is more effective in attacking, which significantly improves the efficiency and accuracy of the attack.

(A): Using only the CNN benchmark model can effectively extract features and capture leaked information. However, it does not perform well on the ASCAD_Desnyc50 dataset.

(A + B): Combining the SE module with the convolutional neural network has been verified to be effective in capturing global information and feature extraction capabilities, and better focusing on leaked information.

(A + C): Finally, the hybrid attention mechanism is combined with the benchmark model. Using an improved channel attention mechanism and spatial attention mechanism can effectively capture information and extract feature location and weight information. It performs better on the ASCAD_Desnyc100 dataset and converges faster than other modules.



(a)Accuracy curve for each module

(b)Comparison of the performance of the modules in the ASCAD dataset

Fig. 5 Performance curve of each module

4.3.2 Comparison experiment

In order to verify the validity of the proposed model and strengthen the comparative analysis, in this section, the HACNN model is trained on the ASCAD public dataset to represent the performance of the attention network, and the existing mainstream neural network models are reproduced. By comparing the CNNbest model, the lightweight network model proposed by Zaid et al., and the HACNN model in this paper, the following comparative conclusions are drawn, which show that our model is superior to previous work.

The results of the experiment under the ASCAD.h5 dataset are shown in Fig. 6a. In the experiment in this paper, the power consumption curve to be tested is randomly selected from the test dataset for each attack. The Rank metric is evaluated and compared for these two networks, and the comparison shows the efficient performance of the model in this paper. Table 3 shows the number of energy traces required to successfully attack the model. The CNN best model proposed by Prouff et al. [27] requires 510 energy traces before the rank metric slowly and steadily decreases to 0. The model proposed by Zaid et al. [28] requires only 191 energy traces to successfully attack the model. In contrast, the HACNN model proposed in this paper requires only 48 energy traces to successfully attack the model on this dataset. This verifies the performance of the hybrid attention network proposed in this paper on this dataset and the superior attack results compared with those in the literature [27] and the literature [28].

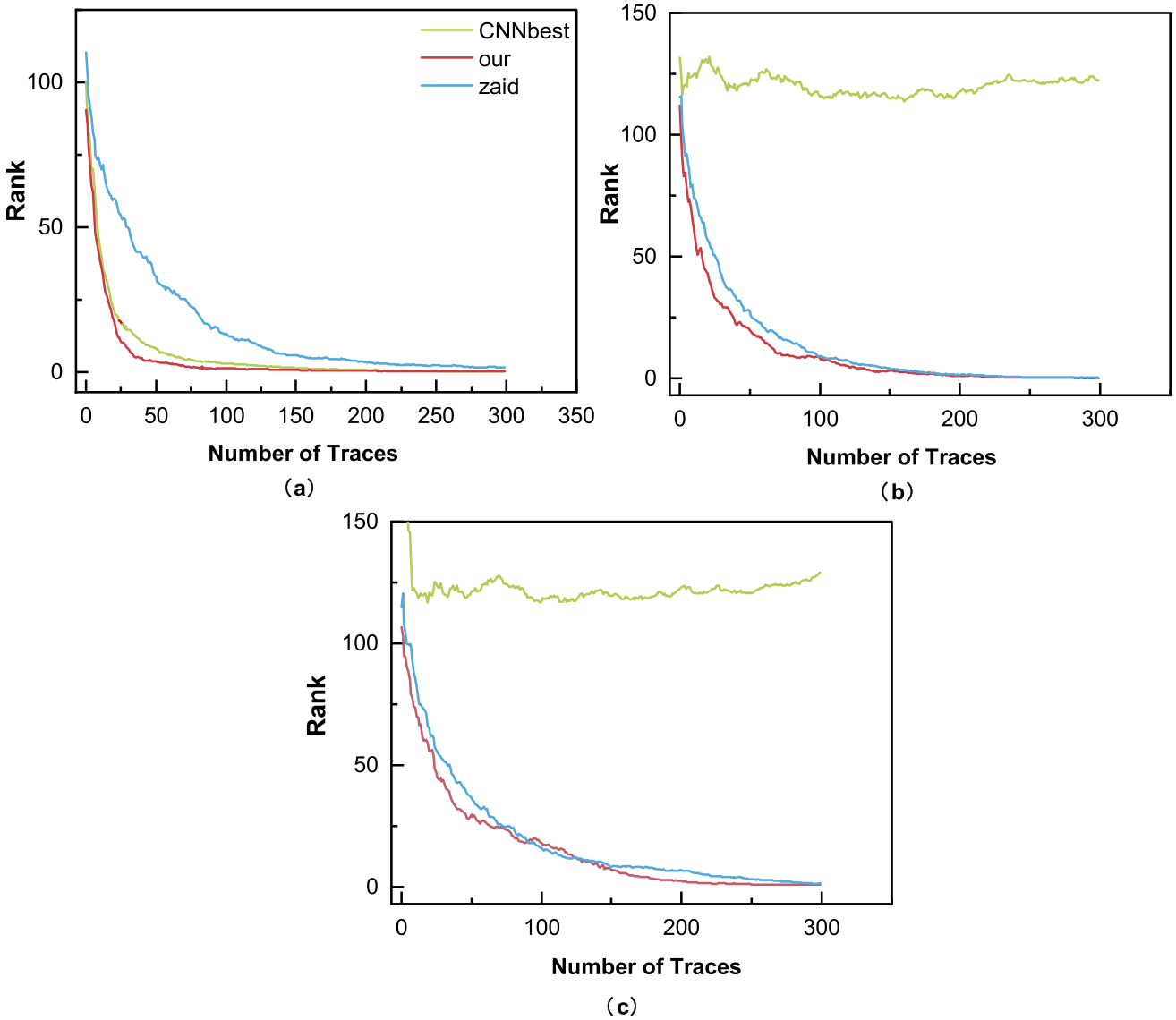


Fig. 6 **a** Average rank in ASCAD. **b** Average rank in ASCAD_Desnyc50. **c** Average rank in ASCAD_Desnyc100

Table 3 Comparison of attack effects on ASCAD dataset

Method	ASCAD_Desnyc0	ASCAD_Desnyc50	ASCAD_Desnyc100
CNNbest [27]	510	> 5000	> 5000
Zaid [28]	191	244	270
Our	48	90	155

When the target dataset is ASCAD_desync50, the experimental results are shown in Fig. 6b. As can be seen in Fig. 6b, the best of the CNN models in [27] cannot recover the correct key using 5000 energy traces, while the model used in [28] requires only 244 energy traces to succeed in the attack. In contrast, the HACNN model in this paper only needs 90 energy traces to rank the correct key first in a dataset with a maximum asynchronous amount of 50 sample points.

The experimental results on the ASCAD_Desnyc100 dataset are shown in Fig. 6c. As can be seen from Fig. 6c, the HACNN model in this paper only needs 155 energy traces to recover the correct key. The Zaid model [28] needs 270 energy traces to recover the key, while the CNN best model in [27] performs even worse on the ASCAD_Desnyc100 dataset. The test results show that the model in this paper can effectively learn features with the help of a hybrid attention mechanism on the ASCAD dataset, and more efficiently complete key recovery with a minimum number of energy traces.

As shown in Table 3, the CNNbest model, which relies on fixed convolutional kernels to capture features, requires more than 5000 energy trajectories to implement a side-channel attack on the encryption algorithm. Zaid et al.'s improved lightweight model has reduced model complexity by using only one convolutional block and minimizing the filter size, and the number of energy trajectories has been significantly improved compared with the CNNbest model, but it is difficult to implement an attack in the presence of noise. Our model uses a fused attention mechanism that uses a collaborative mechanism of dual attention in both channel and space, which significantly improves efficiency when the number of attack trajectories is the same.

As shown in the table, the attack performance of our model is 90.6% higher than that of the CNNbest model, and the efficiency is 74.9% higher than that of the Zaid model. Moreover, under the same conditions, compared with other models, it only needs 48 curves to successfully attack, which verifies the screening ability of the spatio-temporal attention mechanism. In Desync50, that is, when the time offset is 50, the number of curves required for the model in this paper to successfully attack is reduced by 2.71 times compared with the Zaid model, and it is reduced by 55 times compared with the CNNbest model. For an offset of 100, the HACNN model in this paper requires fewer curves to recover the correct key, using nearly 1.71 times fewer curves than the Zaid model and being nearly 32 times more efficient than the CNNbest model. This is because the model in this paper uses channel weights to suppress noisy channels, spatial weights to eliminate redundant sample points, and adaptive capture of offset features to solve the shortcomings of other models, achieving a qualitative leap in attack efficiency under highly desynchronized scenarios.

5 Conclusions

In this paper, a convolutional neural network architectural model for side channel attacks is proposed, which aims to effectively improve the efficiency of neural network-based side channel modelling and analysis methods, and is capable of extracting feature information from energy trace curves to achieve side channel attacks. By using a convolutional module to extract features at different scales, and designing a feature extraction module with a hybrid attention mechanism, which consists of a fusion of a channel spatial attention mechanism and a squeeze-and-excite network, it is able to dynamically extract deep features. The input power consumption curve ground features are used for weighting and rescaling to effectively capture the leakage information. Experimental results show that the HACNN model can optimize the feature extraction network, suppress noisy features, improve the learning effect of key features, and recover the key using fewer energy traces. Only 48 energy trace curves are needed to successfully attack the synchronous dataset. Compared with existing models, the key can be recovered using fewer power consumption energy trace curves, and the attack performance is significantly improved. The model accuracy is increased to 98.12%, and the efficiency is increased by an average of 74.9% compared with existing models.

However, considering that current research does not systematically optimize parameters and that the masking measures in the actual situation result in greater noise interference in actual usage scenarios, the model may face the risk of performance degradation. Although the improved attention mechanism improves the allocation of feature weights,

it does not quantitatively analyze the convergence speed and resource consumption of the model, and lacks a visual explanation of the focusing process on key feature points, which poses the risk of black-box optimization. Therefore, future research directions can further explore the specific impact of attention mechanisms and the optimization of model performance by hyperparameters and training strategies. Construct diverse datasets that include noise and protection mechanisms, study attention mechanism models or other model architectures that are suitable for noisy datasets, and improve the stability of models in real attack scenarios. Research lightweight dynamic attention modules for side channel attacks and design model architectures for lightweight designs. Balance computational efficiency and feature selection capabilities through pruning and quantization techniques to meet the deployment requirements of edge devices. Introduce gradient-weighted activation mapping to further study model interpretability. In the future, attack models can also be integrated into an automated side channel assessment system, combining the entire process of power consumption collection, feature preprocessing, and key recovery to achieve quantitative assessment of security vulnerabilities and generate repair recommendations.

Author contributions Tao Feng: Conceptualization, methodology, formal analysis, funding acquisition, project administration, resources, validation, writing review and editing. Huan Gao: Investigation, methodology, supervision, conceptualization, data curation, resources, software, validation, writing original draft. Xiaomin Li: Resources, software, supervision, data curation, methodology. Chunyan Liu: Data curation, investigation, software, supervision, validation.

Funding This work is supported by the National Natural Science Foundation of China (Grant Nos. 62162039, 61762060, 62462044), Natural Science Foundation of Gansu Province, China (23YFGA0060, 23JRRA837, 24YFFA016).

Availability of data and materials The datasets generated during and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval and consent to participate This article does not contain any studies with human participants performed by any of the authors.

Consent for publication Consent for publication was obtained from the participants.

Competing interests The authors declare that they have no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Vaiyapuri T, Shankar K, Rajendran S, et al. Blockchain assisted data edge verification with consensus algorithm for machine learning assisted IoT. IEEE Access. 2023;11:55370–55379.
2. Fukuda Y, Yoshida K, Fujino T. CA-SCA: non-profiled deep learning-based side-channel attacks by using cluster analysis. IEICE Transactions on fundamentals of electronics, communications and computer sciences. 2024.
3. Vaiyapuri T, Shankar K, Rajendran S, et al. Automated cyberattack detection using optimal ensemble deep learning model. Trans Emerg Telecommun Technol. 2024;35(4):e4899.
4. Duan J, Xiong J, Li Y, et al. Deep learning based multimodal biomedical data fusion: An overview and comparative review. Inf Fusion. 2024;102536.
5. Chauhan VK, Zhou J, Lu P, et al. A brief review of hypernetworks in deep learning. Artif Intell Rev. 2024;57(9):250.
6. Shi Z, Mamun A A, Kan C, et al. An LSTM-autoencoder based online side channel monitoring approach for cyber-physical attack detection in additive manufacturing. J Intell Manuf. 2023;1–17.
7. Hu T, Xu C, Zhang S, et al. Cross-site scripting detection with two-channel feature fusion embedded in self-attention mechanism. Comput Secur. 2023;124:102990.
8. Xiao C, Tang M, Guilley S. Exploiting the microarchitectural leakage of prefetching activities for side-channel attacks. J Syst Archit. 2023;139:102877.

9. Tanaka Y, Ueno R, Xagawa K, et al. Multiple-valued plaintext-checking side-channel attacks on post-quantum kems. IACR Trans Cryptogr Hardw Embed Syst. 2023;2023(3):473–503.
10. Kocher P, Jaffe J, Jun B. Differential power analysis. In: Advances in cryptology—CRYPTO'99: 19th Annual international cryptology conference Santa Barbara, California, USA, August 15–19, 1999 proceedings 19. Springer; 1999, pp. 388–397.
11. Li L, Yu Ou. A deep learning-based side channel attack model for different block ciphers. J Comput Sci. 2023;72:102078.
12. Kumar KN, Mohan CK, Cenkeramaddi LR. The impact of adversarial attacks on federated learning: a survey. IEEE Trans Pattern Anal Mach Intell. 2023;46(5):2672–91.
13. Abbas S, Alsubai S, Ojo S, et al. Active learning for detecting hardware sensors-based side-channel attack on smartphone. Arab J Sci Eng. 2025;50(2):877–89.
14. Alabdulwahab S, Cheong M, Seo A, et al. Enhancing deep learning-based side-channel analysis using feature engineering in a fully simulated IoT system. Expert Syst Appl. 2025;266:126079.
15. Dang S, Shao Y, Li Z, et al. SAluMC: thwarting side-channel attacks via random number injection in RISC-V. Entropy. 2025;27(2):202.
16. Ni C, Li SC. Machine learning enabled industrial IoT security: challenges, trends and solutions. J Ind Inf Integr. 2024;100549.
17. Rehman A, Razzak I, Xu G. Federated learning for privacy preservation of healthcare data from smartphone-based side-channel attacks. IEEE J Biomed Health Inform. 2022;27(2):684–90.
18. Maghrebi H, Portigliatti T, Prouff E. Breaking cryptographic implementations using deep learning techniques. In: Security, privacy, and applied cryptography engineering: 6th international conference, SPACE 2016, Hyderabad, India, December 14–18, 2016, proceedings 6. Springer; 2016, pp. 3–26.
19. Hospodar G, Mulder ED, Gierlichs B, et al. Least squares support vector machines for side-channel analysis. Center for Advanced Security Research Darmstadt; 2011. p. 99–104.
20. Timon B. Non-profiled deep learning-based side-channel attacks with sensitivity analysis. In: IACR transactions on cryptographic hardware and embedded systems; 2019, pp. 107–131.
21. Benadjila R, Khati L, Vergnaud D. Secure storage—confidentiality and authentication. Comput Sci Rev. 2022;44:100465.
22. Alam M, Bhattacharya S, Mukhopadhyay D. Victims can be saviors: a machine learning–based detection for micro-architectural side-channel attacks. ACM J Emerg Technol Comput Syst (JETC). 2021;17(2):1–31.
23. Ahmed A A, Hasan M K, Memon I, et al. Secure AI for 6G mobile devices: deep learning optimization against side-channel attacks. IEEE Trans. Consum. Electron. 2024.
24. Ito A, Saito K, Ueno R, et al. Imbalanced data problems in deep learning-based side-channel attacks: analysis and solution. IEEE Trans Inf Forens Secur. 2021;16:3790–802.
25. Kim J, Picek S, Heuser A, et al. Make some noise. unleashing the power of convolutional neural networks for profiled side-channel analysis. In: IACR transactions on cryptographic hardware and embedded systems; 2019, pp. 148–179.
26. Picek S, Perin G, Mariot L, et al. Sok: deep learning-based physical side-channel analysis. ACM Comput Surv. 2023;55(11):1–35.
27. Benadjila R, Prouff E, Strullu R, et al. Deep learning for side-channel analysis and introduction to ASCAD database. J Cryptogr Eng. 2020;10(2):163–188.
28. Zaid G, Bossuet L, Habrard A, et al. Methodology for efficient CNN architectures in profiling attacks. In: IACR transactions on cryptographic hardware and embedded systems; 2020, pp. 1–36.
29. Hu F, Shen J, Vijayakumar P. Side-channel attacks based on multi-loss regularized denoising autoencoder. IEEE Trans Inf Forens Secur. 2023;19:2051–65.
30. Ahmed AA, Hasan MK, Islam S, et al. Design of convolutional neural networks architecture for non-profiled side-channel attack detection. Elektronika Ir Elektrotechnika. 2023;29(4):76–81.
31. Do NT, Hoang VP, Doan VS, et al. On the performance of non-profiled side channel attacks based on deep learning techniques. IET Inf Secur. 2023;17(3):377–93.
32. Lu X, Zhang C, Cao P, et al. Pay attention to raw traces: a deep learning architecture for end-to-end profiling attacks. In: IACR transactions on cryptographic hardware and embedded systems; 2021, pp. 235–274.
33. Kubota T, Yoshida K, Shiozaki M, et al. Deep learning side-channel attack against hardware implementations of AES. Microprocess Microsyst. 2021;87:103383.
34. Priya SSS, Karthigaikumar P, Teja NR. FPGA implementation of AES algorithm for high speed applications. In: Analog integrated circuits and signal processing; 2022, pp. 1–11.
35. Kumar TM, Reddy KS, Rinaldi S, et al. A low area high speed FPGA implementation of AES architecture for cryptography application. Electronics. 2021;10(16):2023.
36. Ou Y, Li L. Side-channel analysis attacks based on deep learning network. Front Comput Sci. 2022;16(2):162303.
37. Kwon D, Hong S, Kim H. Optimizing implementations of non-profiled deep learning-based side-channel attacks. IEEE Access. 2022;10:5957–67.
38. Alabsi BA, Anbar M, Rihan SDA. CNN-CNN: dual convolutional neural network approach for feature selection and attack detection on internet of things networks. Sensors. 2023;23(14):6507.
39. Chang L, Wei Y, He S, et al. Research on side-channel analysis based on deep learning with different sample data. Appl Sci. 2022;12(16):8246.
40. Zamanzadeh Darban Z, Webb GI, Pan S, et al. Deep learning for time series anomaly detection: a survey. ACM Comput Surv. 2024;57(1):1–42.
41. Woo S, Park J, Lee J Y, et al. CBAM: convolutional block attention module. In: Proceedings of the European conference on computer vision (ECCV); 2018, pp. 3–19.
42. Al-Hchaimi AAJ, Sulaiman NB, Mustafa MAB, et al. A comprehensive evaluation approach for efficient countermeasure techniques against timing side-channel attack on MPSoC-based IoT using multi-criteria decision-making methods. Egypt Inform J. 2023;24(2):351–64.
43. Li L, Ou Y. A deep learning-based side channel attack model for different block ciphers. J Comput Sci. 2023;72:102078.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.